Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

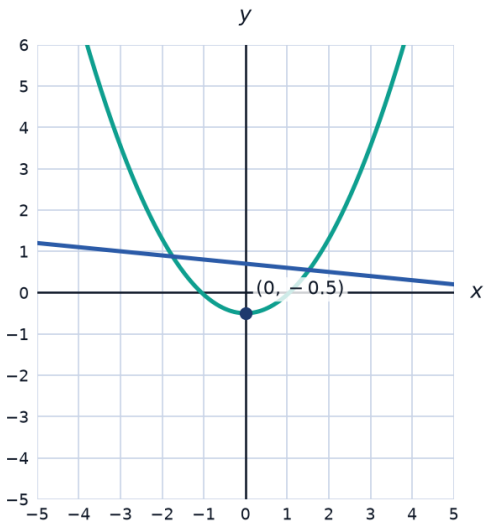
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Complex roots of a quadratic**

Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

2 **[Foundational]** • **Target: Complex roots of a quadratic**

Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.



3 **[Foundational]** • **Target: Modulus**

Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

4 **[Foundational]** • **Target: Modulus**

Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

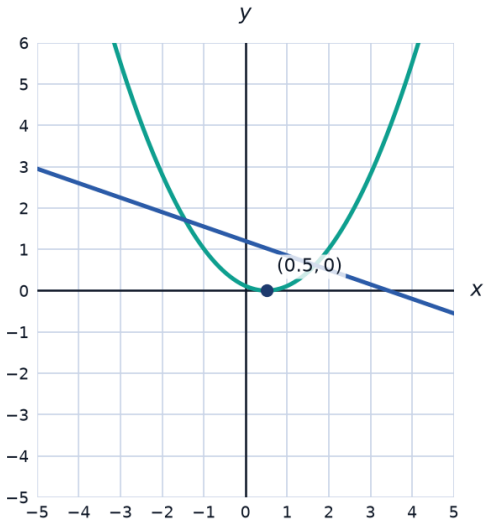
5 **[Foundational]** • **Target: Complex roots of a quadratic**

Evaluate the expression $\frac{(3+2i)^2}{i}$ and write the result in standard form $a + bi$.

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6 [Foundational] • Target: Modulus

[Complex Numbers · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.



7 [Foundational] • Target: Arithmetic with complex numbers

Solve the quadratic equation $2x^2 - 6x + 9 = 0$ over the complex numbers, and state the sum and product of the roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Complex roots of a quadratic

An arithmetic sequence has third term $a_3 = 11$ and eighth term $a_8 = 31$. Find the common difference d, the first term a_1, and the sum of the first 20 terms S_{20}.

9 [Intermediate] • Target: Modulus

Let $z = \frac{5+2i}{2-3i} + i^{43}$. (a) Express z in standard form $a + bi$. (b) Compute the modulus $|z|$.

10 [Intermediate] • Target: Arithmetic with complex numbers

Express $z = \frac{4+7i}{1-2i}$ in standard form $a + bi$, and compute its modulus squared $|z|^2$.

11 [Intermediate] • Target: Complex roots of a quadratic

The sum of the first n terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term a_{10}.

12 [Intermediate] • Target: Modulus

Given $f(x) = \frac{2x+5}{x-3}$ for $x \neq 3$, find the inverse function $f^{-1}(x)$ and state its domain.

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13 [Intermediate] • Target: Arithmetic with complex numbers

A committee of 4 people is to be selected from a group of 6 men and 5 women. What is the probability that the committee consists of exactly 2 men and 2 women?

14 [Intermediate] • Target: Complex roots of a quadratic

Let $A = (4, 3; 2, 2)$. Find A^{-1} and compute the vector $\mathbf{x} = A^{-1}(10; 6)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Modulus • Cross-Lesson: 6.01 Probability

A student answers 35 questions in 42 minutes. To finish all 50 questions on the exam, how many seconds per question must the student maintain for the remaining questions?

16 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 2.01 Sequences & Series

Find the equation of the tangent line to the parabola $f(x) = 2x^2 - 3x + 1$ at the point where $x = 2$.

17 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 3.01 Functions & Graphs

[Test format & strategy · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

18 [Challenging] • Target: Modulus • Cross-Lesson: 4.01 Matrices & Vectors

[Functions & Graphs · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

19 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 5.01 Introductory Calculus

[Introductory Calculus · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

20 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 6.01 Probability

[Matrices & Vectors · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Arithmetic with complex numbers

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

a. 6

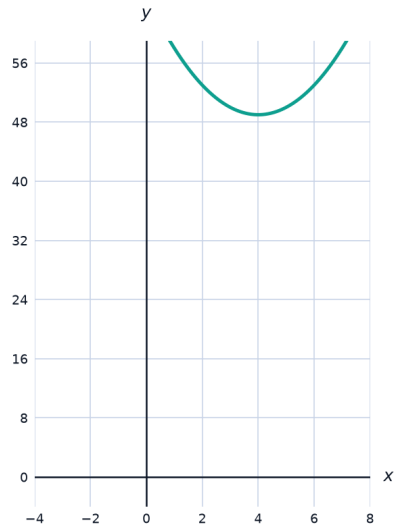
b. 8

c. 10

d. 12

22 [Foundational] • Target: Complex roots of a quadratic

Find the product of the roots of $x^2 - 8x + 65 = 0$.



a. 65

b. 67

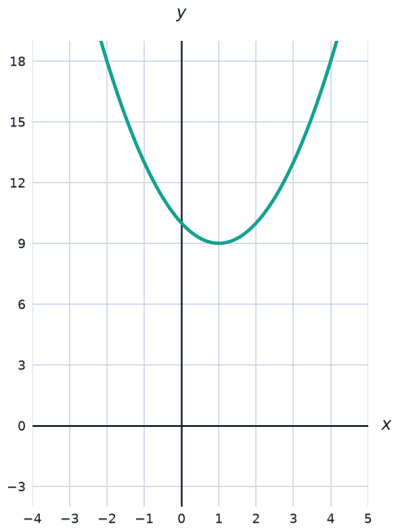
c. -8

d. -65

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23 [Foundational] • Target: Modulus

Find the product of the two roots of $x^2 - 2x + 10 = 0$.



a. 10

b. 1

c. -9

d. 2

24 [Foundational] • Target: Arithmetic with complex numbers

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. 5

b. 7

c. 12

d. 25

25 [Foundational] • Target: Complex roots of a quadratic

For which value of k does $x^2 + kx + 16 = 0$ have complex (non-real) roots?

a. $k = 7$

b. $k = 9$

c. $k = 8$

d. $k = -10$

26 [Foundational] • Target: Modulus

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

a. $2 + 3i$

b. $2 - 3i$

c. $\frac{10+15i}{3}$

d. $\frac{18+15i}{5}$

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27 [Foundational] • Target: Arithmetic with complex numbers

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

a. 5

b. 6

c. 7

d. 8

28 [Foundational] • Target: Complex roots of a quadratic

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

a. $\frac{17}{2}$

b. $-\frac{17}{2}$

c. $\frac{11}{2}$

d. $-\frac{11}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Modulus

Let $z = 3 - 4i$. What is the real part of z?

a. 3

b. -4

c. 5

d. 7

30 [Intermediate] • Target: Arithmetic with complex numbers

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

a. 5

b. 7

c. 12

d. 25

31 [Intermediate] • Target: Complex roots of a quadratic

Solve $x^2 + 10x + 61 = 0$. Which root has positive imaginary part?

a. -5+6i

b. -5-6i

c. 5+6i

d. 5-6i

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32 [Intermediate] • Target: Modulus

Solve $x^2 + 4x + 8 = 0$. Which root has negative imaginary part?

a. $-2 - 2i$

b. $-2 + 2i$

c. $2 - 2i$

d. $-4 - 2i$

33 [Intermediate] • Target: Arithmetic with complex numbers

What is the complex conjugate of $-3 + 7i$?

a. $-3 - 7i$

b. $3 + 7i$

c. $3 - 7i$

d. $-3 + 7i$

34 [Intermediate] • Target: Complex roots of a quadratic

Find the complex conjugate of $-10 + 9i$.

a. -10-9i

b. 10+9i

c. 10-9i

d. -10+9i

35 [Intermediate] • Target: Modulus

What is the real part of $z = -4 + 9i$?

a. -4

b. 9

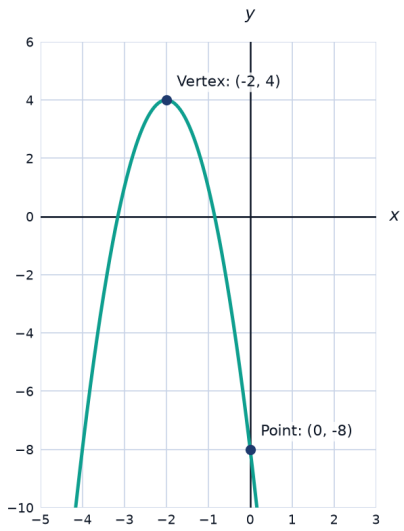
c. 4

d. -9

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36 **[Intermediate]** • **Target: Arithmetic with complex numbers**

Consider the function $g(x) = ax^2 + bx + c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2, 4)$. The parabola passes through the point $(0, -8)$. Find the value of the parameter a.



a. 2

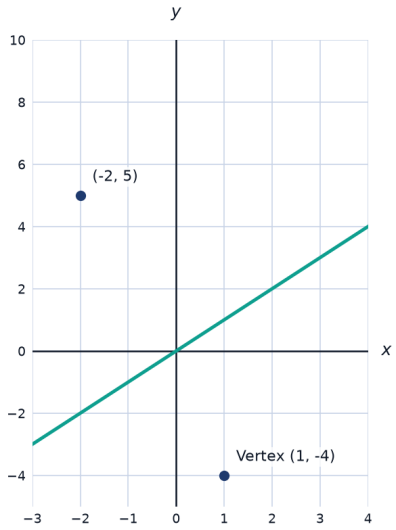
b. -2

c. 4

d. -4

37 **[Intermediate]** • **Target: Complex roots of a quadratic**

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.



a. 1

b. -2

c. -1

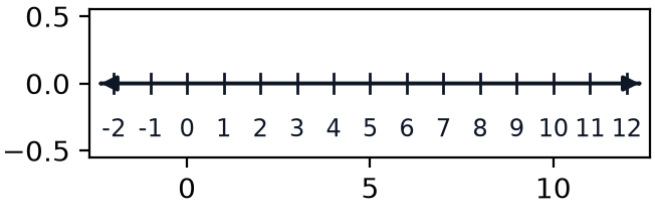
d. 2

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus • Cross-Lesson: 4.01 Matrices & Vectors

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?



a. 3

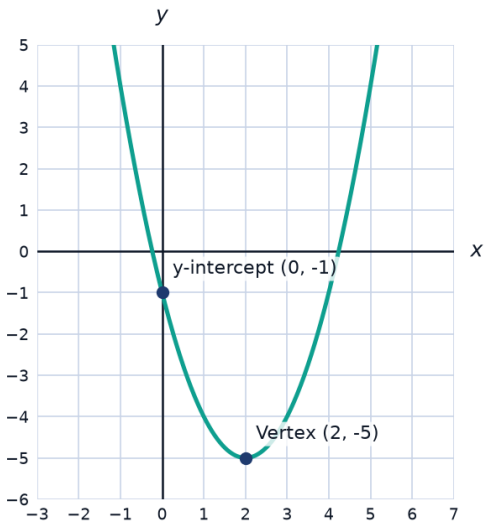
b. -3

c. 12

d. 6

39 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 5.01 Introductory Calculus

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.



a. 1

b. -1

c. 2

d. -2

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40 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 6.01 Probability

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

a. 14

b. 17

c. 8

d. 11

41 [Challenging] • Target: Modulus • Cross-Lesson: 2.01 Sequences & Series

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

a. 17

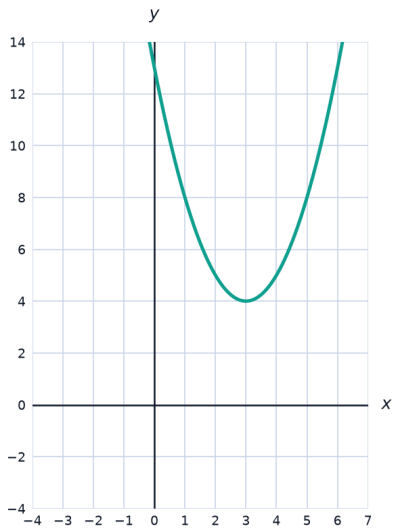
b. 12

c. 21

d. 35

42 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 3.01 Functions & Graphs

Find the sum of the two roots of $x^2 - 6x + 13 = 0$.



a. 6

b. 3

c. 13

d. $4i$

43 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 4.01 Matrices & Vectors

An arithmetic series has $a_1 = 6$, $d = 2$, and $S_n = 204$. Find n.

a. 12

b. 13

c. 10

d. 24

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44 [Challenging] • Target: Modulus • Cross-Lesson: 5.01 Introductory Calculus

An arithmetic sequence has $a_3 = 0$ and $a_7 = 16$. Find a_1.

a. -8

b. -4

c. 8

d. -9

45 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 6.01 Probability

Find the common difference of the arithmetic sequence $6,\ 4,\ 2,\ 0,\ \ldots$

a. -2

b. 2

c. -1

d. 4

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